

Jongmin Choi

M.S. Student in Electrical Engineering · Multimodal AI (MMAI) Lab, KAIST

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RESEARCH INTERESTS

My research asks two questions: how neural networks should compute, and how different modalities should meet inside a single model. Along these lines, I have worked on interpretable spiking neurons for Spiking Neural Networks and on mitigating the audio-text modality gap. I am currently designing connectors that bridge audio encoders and language-model decoders in Audio Language Models, and exploring optimization algorithms for training deep networks.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea Mar 2025 – Present

M.S. in Electrical Engineering

Advisor: **Prof. Joon Son Chung** · Multimodal AI (MMAI) Lab · GPA 4.06 / 4.30

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea Mar 2018 – Feb 2025

B.S. in Electrical Engineering and Mathematical Sciences (Double Major)

GPA 3.65 / 4.30

PUBLICATIONS

[C] Conference · [P] Preprint · * Equal contribution

[P] **FiTS: Interpretable Spiking Neurons via Frequency Selectivity and Temporal Shaping**

Jongmin Choi, Joon Son Chung

arXiv preprint · 2026

[C] **LAMB: LLM-based Audio Captioning with Modality Gap Bridging via Cauchy-Schwarz Divergence**

Hyeongkeun Lee*, **Jongmin Choi***, KiHyun Nam, Joon Son Chung

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) · 2026 **ORAL**

[C] **Diffusion-Link: Diffusion Probabilistic Model for Bridging the Audio-Text Modality Gap**

KiHyun Nam*, **Jongmin Choi***, Hyeongkeun Lee, Jungwoo Heo, Joon Son Chung

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) · 2026 **POSTER**

[P] **Fork-Merge Decoding: Enhancing Multimodal Understanding in Audio-Visual Large Language Models**

Chaeyoung Jung*, Youngjoon Jang*, **Jongmin Choi**, Joon Son Chung

arXiv preprint · 2025

TEACHING EXPERIENCE

EE.70038 — Speech Recognition Systems — KAIST

Spring 2026

Teaching Assistant · Instructor: **Prof. Joon Son Chung**

EE.40034 — Deep Learning for Visual Understanding — KAIST

Fall 2025

Teaching Assistant · Instructor: **Prof. Joon Son Chung**

INDEPENDENT SOFTWARE PROJECTS

PickCam

iOS · App Store

Sole developer of an iOS app that reframes multi-person video around a chosen subject, using on-device person detection, segmentation, and face re-identification (Vision, Core ML, AVFoundation, Metal).

Nureum (누름꽃)

iOS / iPadOS · App Store

Sole developer of an offline-first iOS app that turns photographed flowers into digital pressed-flower keepsakes via on-device segmentation, with a 355-entry guide localized in eight languages (Vision, Core Image, SwiftData, CloudKit).

SKILLS

Research & ML: Python, PyTorch · **iOS Development:** Swift, SwiftUI, SwiftData · **Languages:** Korean (native), English (fluent)

Last updated: August 2026